

Project HERMES

Development of a Small-Scale Rocket and Adaptive Simulation for Propulsive Landings

All graphics, tables, and images have been created by Kabir Singh & Agastya Mishra, 2026 unless otherwise denoted by an asterisk.

Category: Engineering Technology: Statics & Dynamics [Computational Mechanics]

Introduction

What is suborbital flight?

Suborbital flight is a spaceflight in which a vehicle crosses the **Kármán line**, located at 100 km above sea level, but travels at an insufficient horizontal velocity relative to the Earth's surface to stay in orbit. This results in a **ballistic trajectory** that returns to Earth's surface.

Figure 1: Ballistic Trajectory Diagram [1]*

Uses - They're used for atmospheric science (measuring ozone, aerosols, radiation) and microgravity research (materials science, biology, etc.)

The Issue with Suborbital Rockets

Commercial suborbital flights cost hundreds of thousands of dollars, and even small CubeSat rideshares run \$30k-\$90k per mission (SatBase, 2025). For university labs, small startups, and researchers in developing countries, access to suborbital flight is effectively unavailable. Current rockets use liquid rocket engines (LREs) for landing, but they are very expensive, difficult to manufacture, and significantly more dangerous to handle.

There exists another type of propulsion system that fixes all these problems: solid rocket motors (SRMs). However, SRMs have one major flaw: they cannot control their thrust while they're ignited, and it has to be determined during the time of manufacturing.

Engineering Objectives

This project aims to develop a suborbital launch system using SRMs. HERMES aims to create a proof-of-concept vehicle capable of launch a payload >30% of the rocket's total apogee, mimicking and validating a successful ejection system. This project also aims to develop a framework to simulate suborbital rocket landing using a hoverslam technique: calculating an ignition altitude precisely timed to the known flight state such that the vehicle reaches <2 m/s velocity at touchdown. The framework is evaluated against three criteria:

Functionality

- Shows small-scale empirical validation of the suborbital launch system
- Shows computational validation of HERMES simulation framework
- Improved success rates for suicide ignition and landing velocity

Robustness

- Hermes Rocket capable of handling difficult atmospheric conditions
- Hermes Software must include configurable launch conditions
- Launch system and software must be adaptable to different rockets

Affordability

- Hermes Rocket must be provide an low-cost alternative to traditional suborbital launch methods
- Hermes ML Model must meet the previous two criteria while being resource efficient (must run on traditional rocket avionics)

Constraints

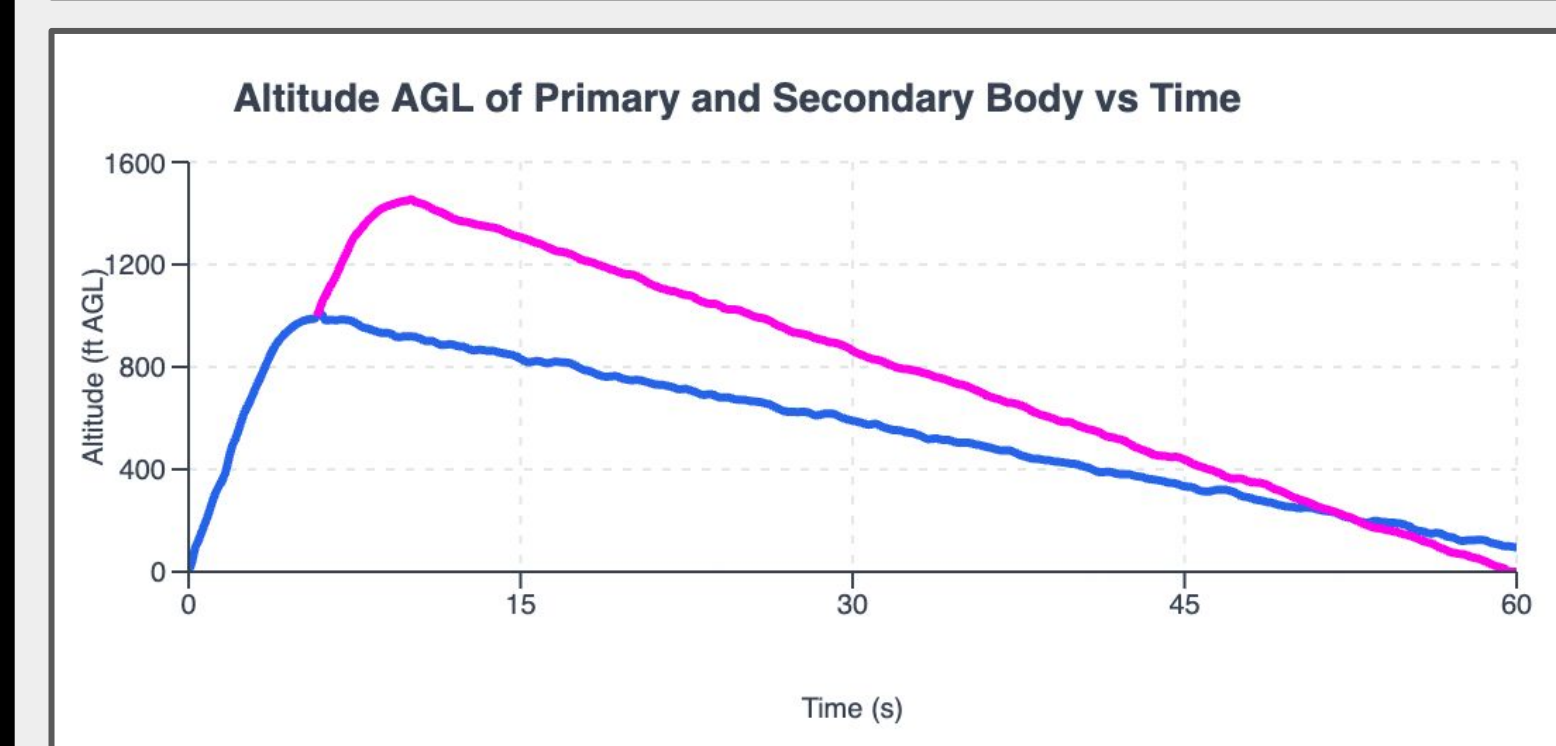
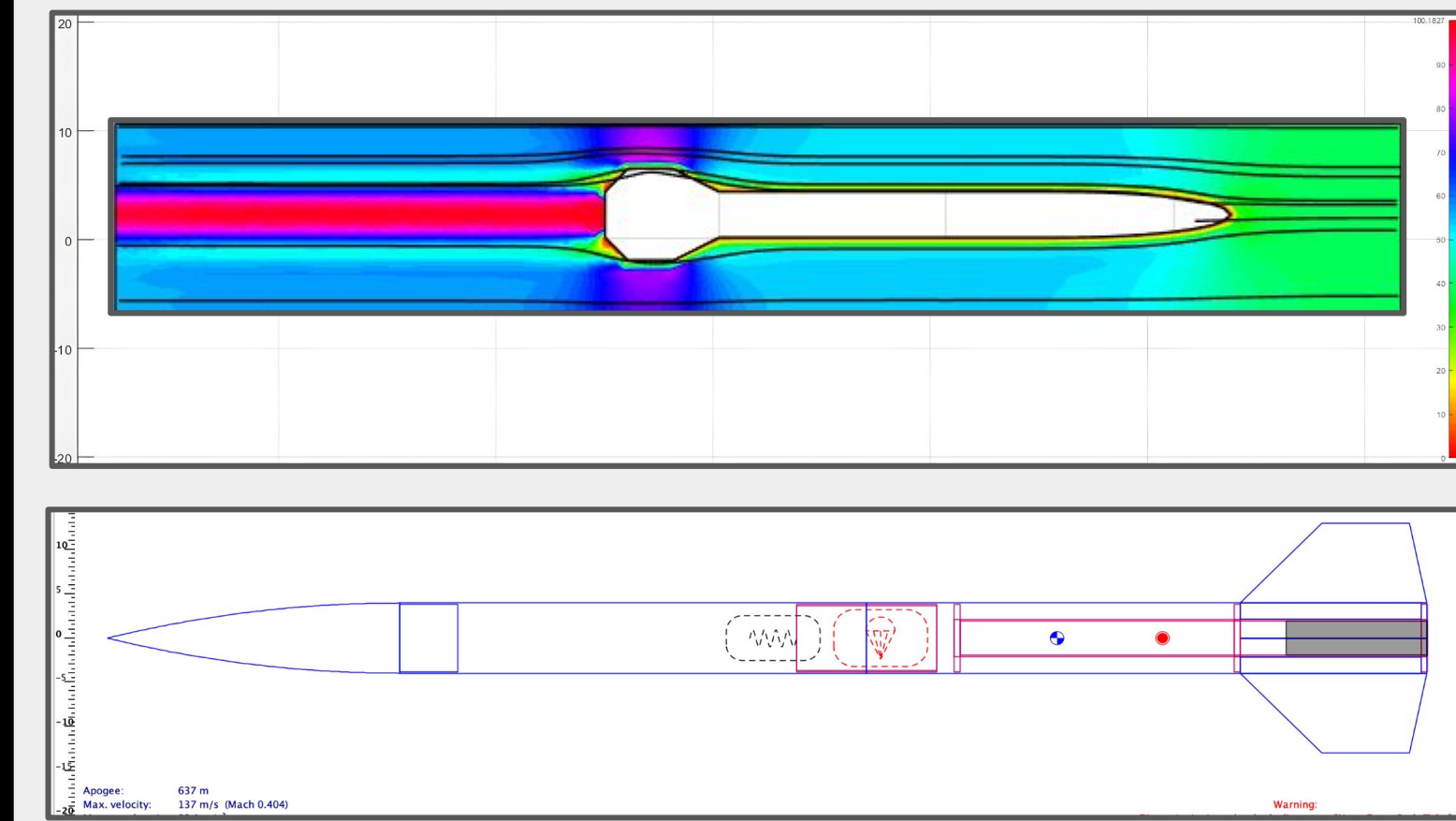
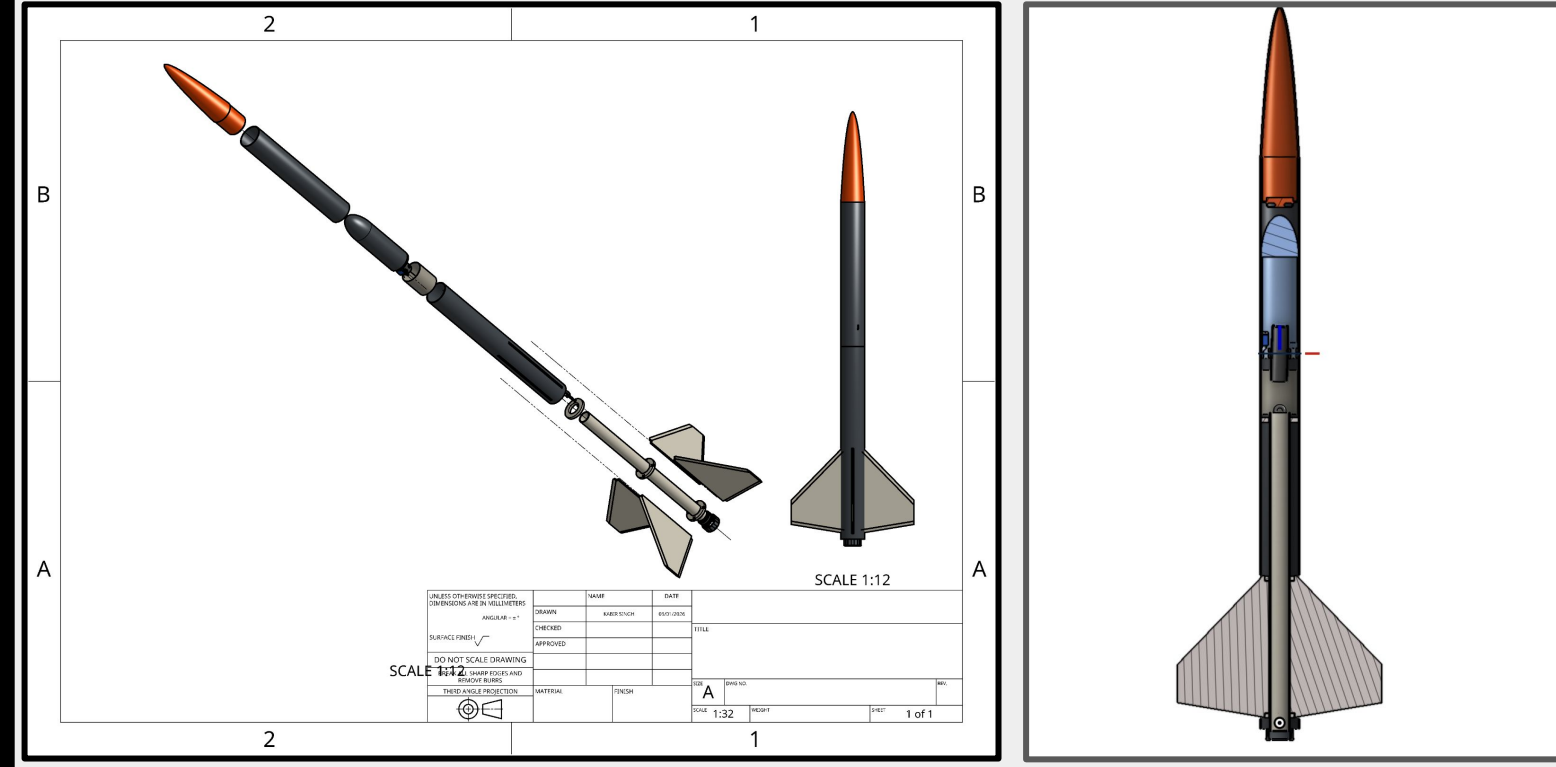
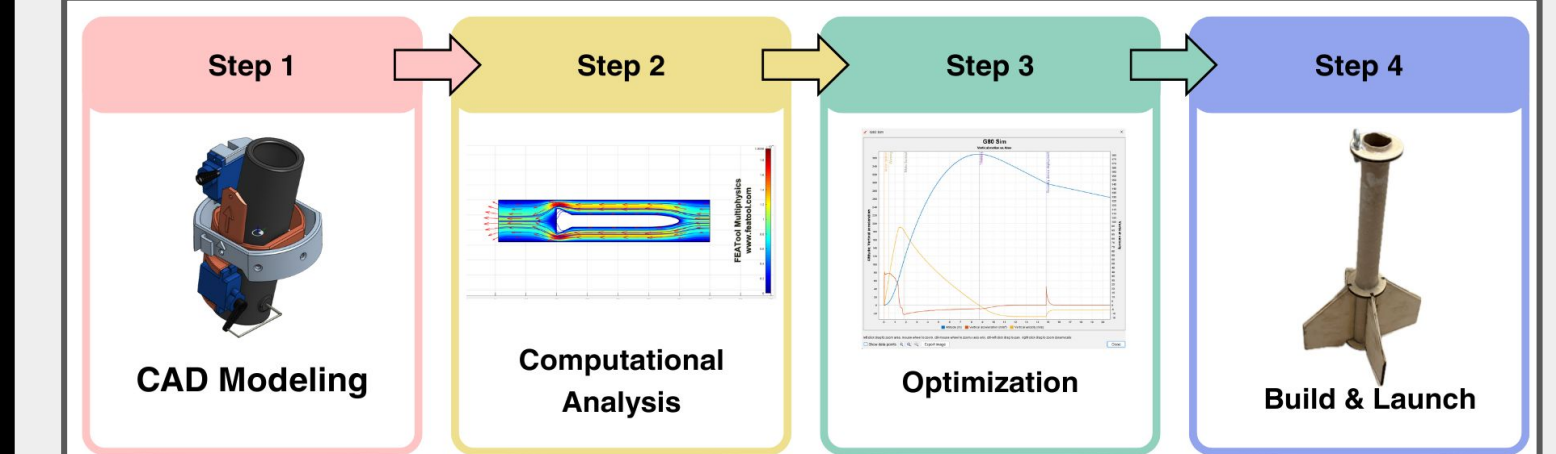
- Logistical
 - Regulatory: NAR/TRA motor certification limits (no L1+ motors without certification)
 - Budget: resources; limited access to professional test facilities
 - Infrastructure: Due to local terrain, rocket launch sites are extremely difficult to locate, and large rocketry associations are scarce within KY
- Technical
 - Motor-to-motor thrust variation
 - Sensor noise: IMU drift, altimeter error
 - Onboard compute: Teensy 4.1 (600 MHz ARM Cortex-M7, 1MB RAM)

Background

- SpaceX (2015) and Blue Origin demonstrated propulsive landing using throttleable liquid engines (Falcon 9 ~40% min thrust); throttle authority is the mechanism that enables real-time correction, which is unavailable in SRMs
- BPS.Space (Barnard, 2022) achieved the first known solid motor propulsive landing through iterative manual tuning across dozens of flights; no systematic optimization methodology was produced or published
- Neural networks have been applied to powered descent guidance in NASA Mars landers and RL-based landing agents, all of which assume throttleable liquid rocket engines with continuous thrust authority. (NASA, 2022)

Supports:	RocketPy	OpenRocket	RockSim
TVC	X	X	X
Landing	X	X	X
Faults & ML	X	X	X
Monte Carlo	✓	✓	✓
Sensor Noise	✓	X	X
Environmental Simulation	✓	✓	✓

Engineering Methods & Design



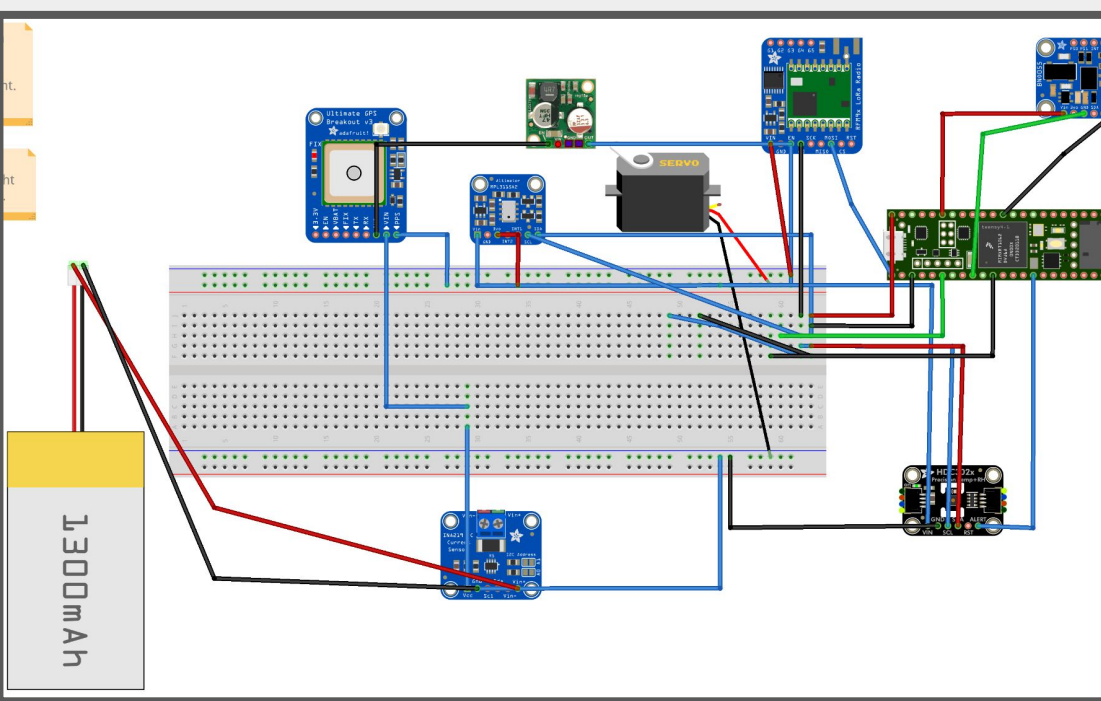
Design Cycle

More than 35 iterative design cycles were conducted over the course of 4 months. Each iteration began with modifying the geometry in CAD to adjust physical parameters.

The updated design was then analyzed using MATLAB FEATool Multiphysics to evaluate aerodynamic characteristics.

While optimizing, a large gap between the scope of HERMES and current modeling tools was identified: standard softwares couldn't account for two-stage launches or compute the change in atmospheric conditions. This led to the creation of the HERMES Simulation Framework.

Insights from the simulations informed the next round of design changes, that allowed us to maximize flight performance before construction of our rocket.



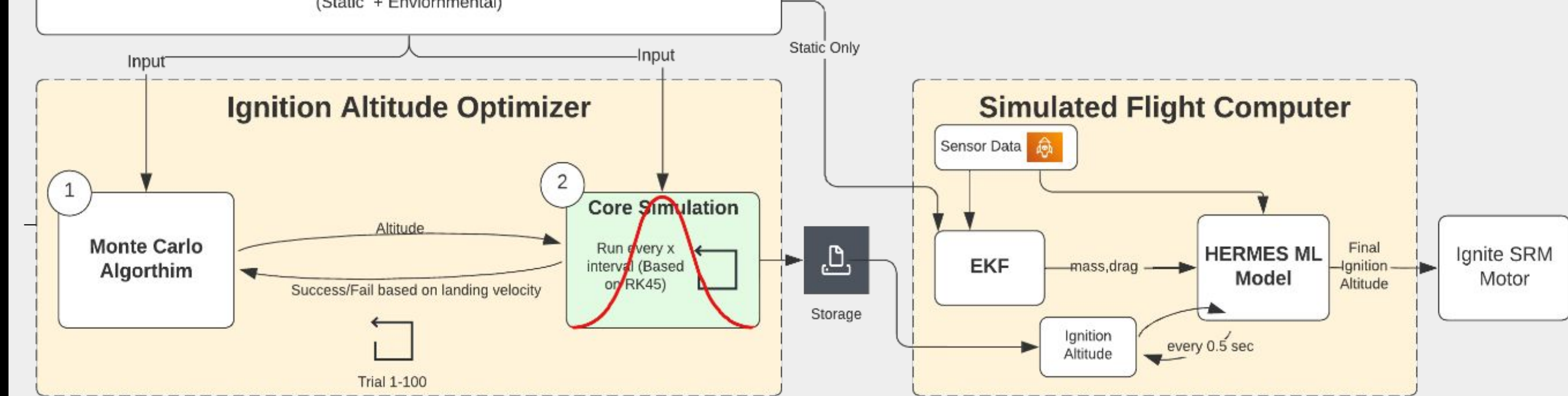
Final Build

Avionics used a BNO055 IMU, MPL3115A2 Barometer, Teensy 4.1 Mainboard, Adafruit GPS, and a RFM95W LoRa Radio. The final build used TotalBoat Thixo Epoxy; when mixed, the epoxy undergoes polymerization and cures into a rigid thermoset plastic with a hardness of 80 Shore D, providing resistance to vibration and stress during launch conditions.

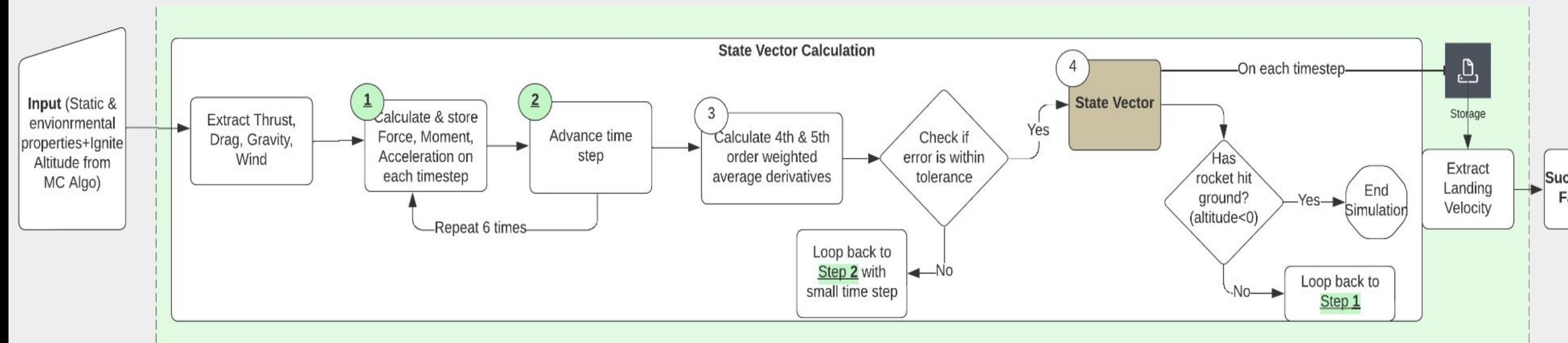
Launch Results

Three independent test launches were conducted to evaluate the rocket's staging system. During each flight, the secondary body was successfully deployed at apogee, demonstrating reliable separation and ignition timing. As a result, the secondary body's propulsion increased the rocket's total achieved apogee by **42.7%**, validating the proof-of-concept for functionality of two-stage payload deployment in suborbital vehicles.

HERMES Framework Overview



Core Simulation



Ignition Altitude Optimizer

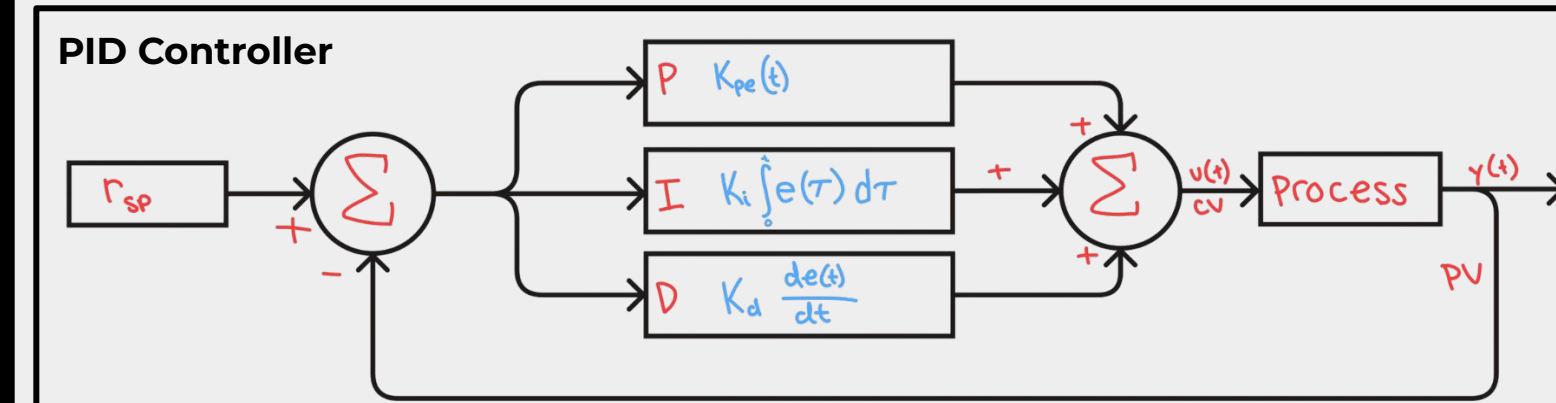
- The optimizer invokes the core simulation engine, passing in a candidate ignition altitude and a sampled parameter set, and receiving back the final flight state.
- This sweeps across 200 candidate altitudes with 100 trials each, independently perturbing thrust, drag, air density, mass flow, and altimeter readings per trial.
- A trial is classified as successful if the vehicle achieves vertical velocity <2 m/s and total velocity <3 m/s.
 - Aggregating across all trials at each altitude produces a success landscape from which the optimal ignition altitude is extracted.

Alt (m)	Trial 1	Trial 2	Trial 3	...	Trial 200	Success Rate
44.8	✓	✗	✗	...	✗	92%
44.9	✓	✓	✗	...	✗	73%
45.0	✓	✓	✓	...	✗	94%
45.1	✗	✓	✓	...	✓	85%
45.2	✓	✓	✓	...	✓	80%

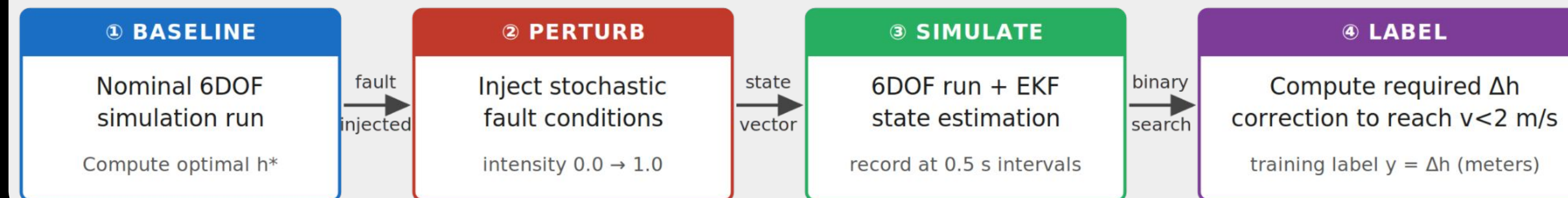
ML Model

Why? The Monte Carlo optimization produces the best fixed ignition altitude (h*). But in flight, actual conditions deviate from pre-computed expectations, and the ~0.3m vertical window leaves minimal margin for error. Every 0.5s during descent, the model reads 19 flight-state features and outputs a scalar correction in meters to the baseline ignition altitude. This corrected altitude is sent to the flight computer, which updates the ignition command in real time.

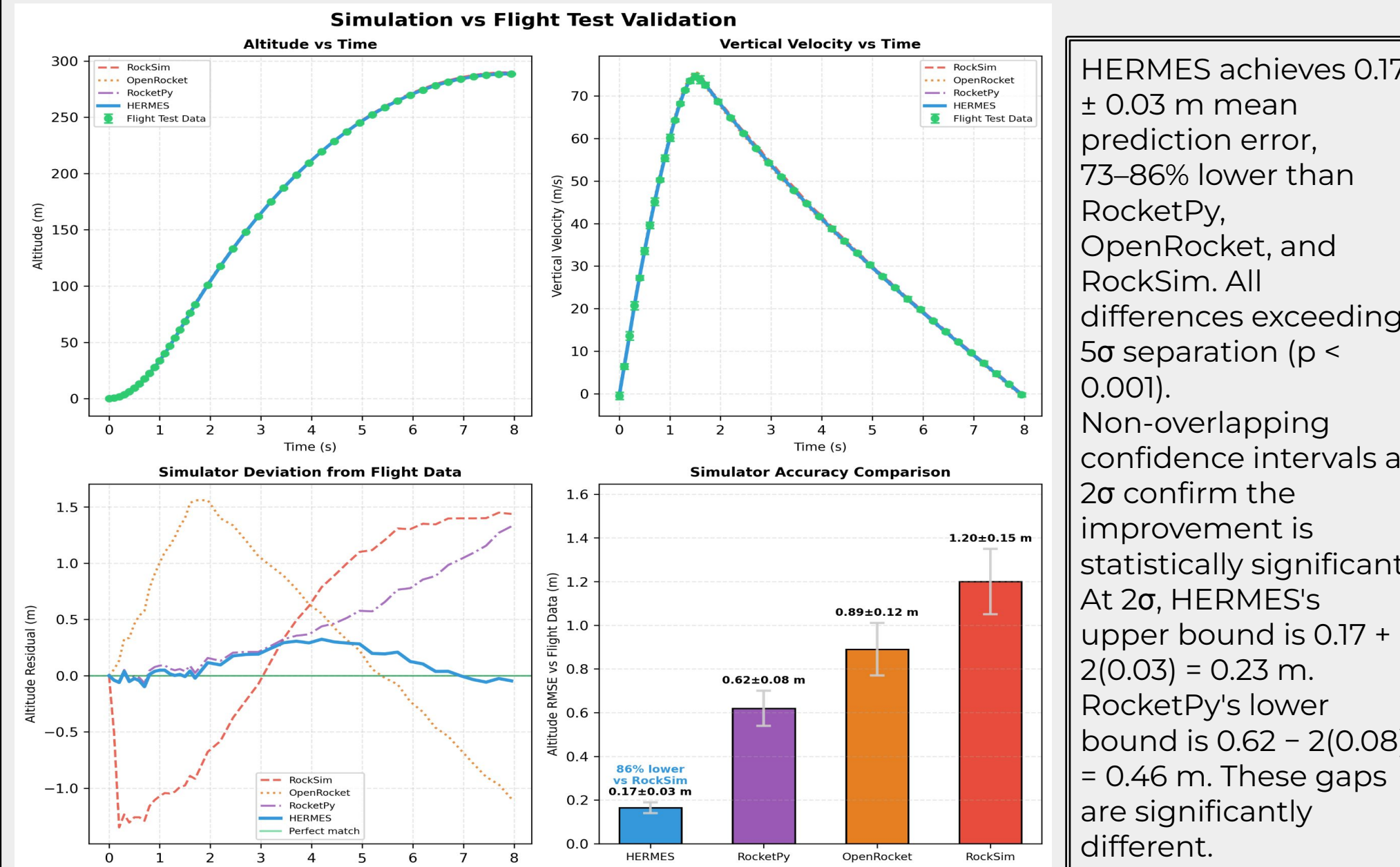
Fault Injection: Introduces parametric disturbances into the simulation to stress-test the guidance algorithm against off-nominal conditions. Faults are applied as perturbations to motor thrust, vehicle mass, drag coefficient, etc. This allows the simulation to evaluate guidance performance across a controlled range of disturbance severities.



Training Data Generation:

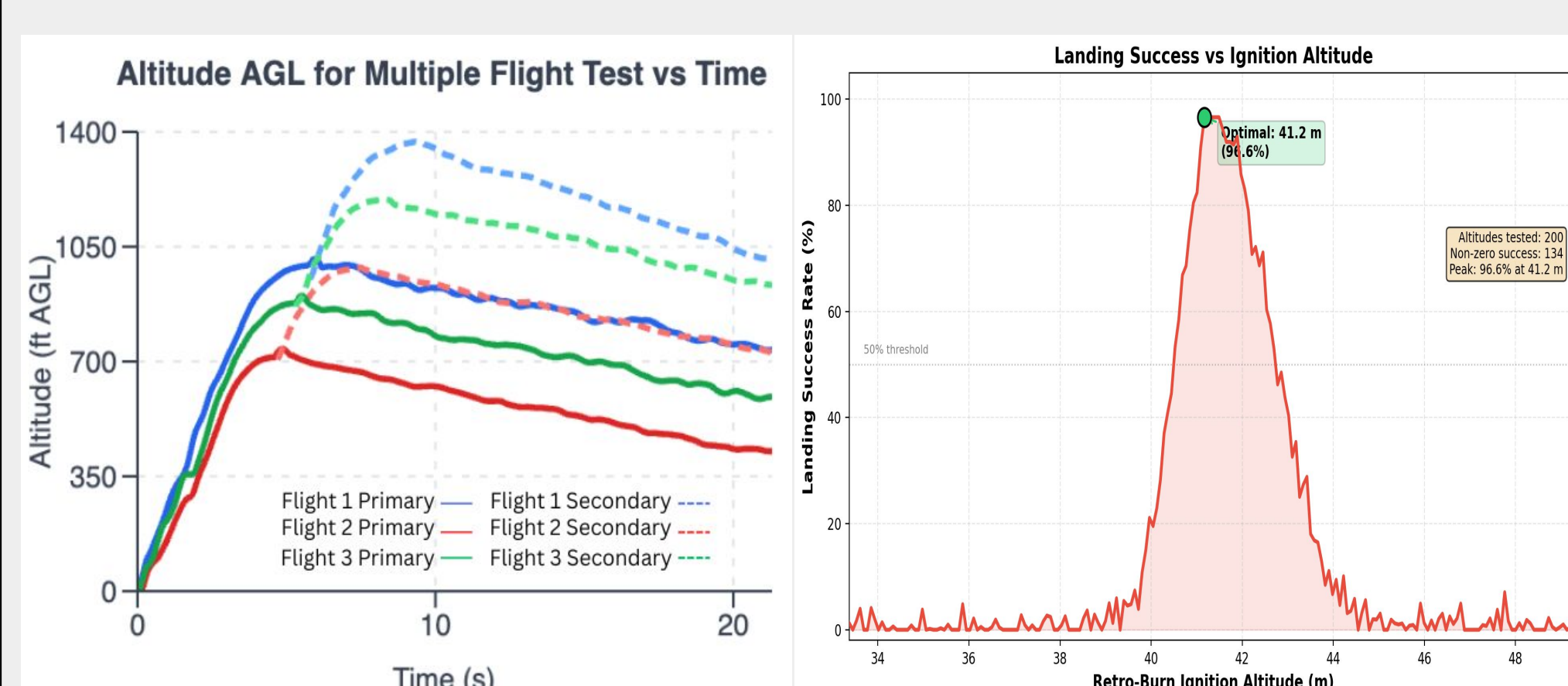


Results & Analysis



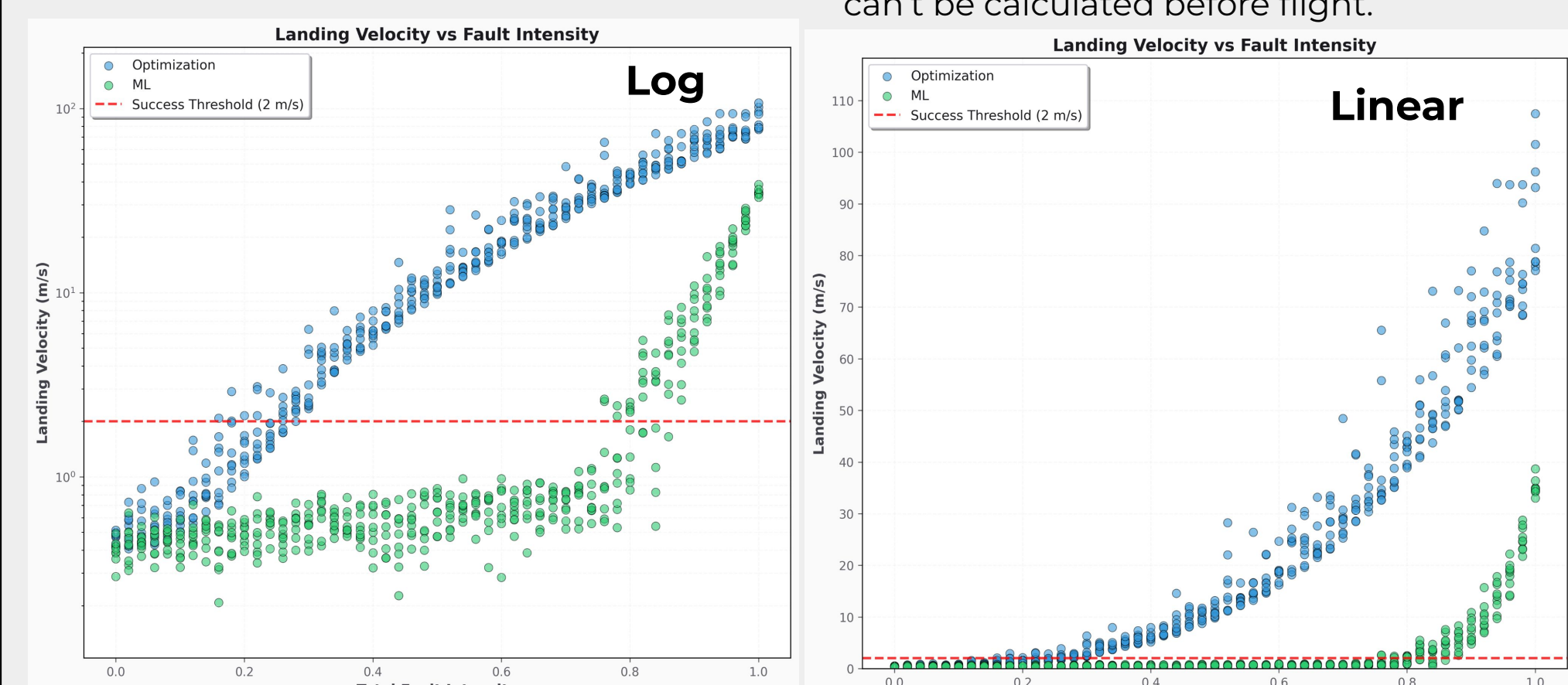
HERMES achieves 0.17 ± 0.03 m mean prediction error, 73-86% lower than RocketPy, OpenRocket, and RockSim. All differences exceeding 5 σ separation ($p < 0.001$). Non-overlapping confidence intervals at 2 σ confirm the improvement is statistically significant. At 2 σ , HERMES's upper bound is $0.17 + 2(0.03) = 0.23$ m. RocketPy's lower bound is $0.62 - 2(0.08) = 0.46$ m. These gaps are significantly different.

The simulation was compared against onboard sensor data from a physical flight test. Where commercially available simulators showed significant deviation from measured flight data, HERMES matched the recorded altitude and velocity profiles to within 0.17m. This accuracy is what qualifies the simulation as a reliable development platform, every subsequent result in this project depends on the simulation behaving like the real rocket.



This graphs shows all three empirical flight tests, along with the trajectory of the deployed payload. Each flight was launched with a different motor in order to test adaptive capability across various configurations.

This curve plots landing success rate against ignition altitude. There is a sharp, **narrow** peak where success is highest. The width of this peak roughly ± 0.3 meters, defines the margin for error, showing that pre-flight computation is insufficient and real-world uncertainty can't be calculated before flight.



Each dot represents one simulated landing. The horizontal axis shows how harsh the conditions were; the vertical axis shows impact speed. The red dashed line marks the 2 m/s survival threshold. For the ML model (green), successful landings form a dense band near 0.5 m/s across all but the most extreme fault levels. The baseline (blue) shows a continuous upward trend: as conditions worsen, impact speed climbs steadily, reaching 60+ m/s at the highest disturbance level. The linear-scale view shows the full range; the log view provides detail near the threshold where the distinction matters most for vehicle survival.

Conclusions

Validation of Engineering Criteria:

- Functionality**
 - 42.7%** greater total achieved apogee, validating suborbital launch concept
 - 73-86% significant** improvement in HERMES Framework accuracy compared to existing softwares
- Robustness**
 - Physical rocket built around **L2 certification** standards and **significantly** higher altitudes
 - 12** Adjustable conditions for Environmental Parameters
 - 12** Factors for Fault Injection
 - 21** Adjustable conditions for Rocket Configuration
- Affordability**
 - BOM estimates **~\$3,700** full-scale build
 - BOM estimates **~\$465** per launch

Takeaways:

HERMES provides a novel foundation for the future of suborbital rockets. Coupled with the HERMES Simulation Framework and Adaptive ML Model, the prototypes validates all engineering criteria while allowing for development in future years.

Future Work

Due to certification constraints, HERMES was unable to complete a higher altitude launch. In the future, HERMES has a set plan for its long term mission of a space shot. This requires:

- Design of a custom SRM propellant - Preliminary Formula:
 - 35.00% Ammonium perchlorate, 400 micron
 - 35.00% Ammonium perchlorate, 90 micron
 - 12.00% Aluminum, 5 micron
 - 1.00% Oxamide, 4 micron
 - 17.00% HTPB binder system
- Fiberglass exterior for high-altitude shots
- Rocket Fin Redesign with Adjustable Axis
- L1/L2 Certification - In correspondence with Nevada Rocketry Club
- ANSYS CFD Analysis
 - Heat Transfer within Motor Tube